

1. Course number and name

CHE 311. Transport Phenomena I

2. Credits, contact hours and categorization

Credit hours: 3; Contact Hours: 150 min/week lecture; Engineering

3. Instructor's or course coordinator's name

Korosh Torabi

4. Text book, title, author, publisher, year

R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2nd Ed., Wiley (2006)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 205, CHE 210, and MATH 220.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Performance indicators of student learning.

- Use mathematical language to perform vector and tensor operations relevant to transport phenomena.
- Comprehend the fundamental physical principles of momentum transport and apply the principle of conservation of momentum to solve basic fluid dynamics problems.
- Set up the differential momentum balance and specify its boundary conditions for various flow problems.
- Solve ordinary differential equations and apply boundary conditions to determine steady-state one-dimensional flow velocity and stress profiles.
- Solve partial differential equations using the methods of separation of variables and combinations of variables to determine the velocity profiles of flow fields with two independent variables.
- Comprehend the physical significance and mathematical formulation of empirical expressions for turbulent momentum transport.
- Understand the definition of friction factors to solve various flow problems.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Week	Topics	Book Chapter
1	Conservation laws, flux, and balance equations Math Review	Chapter 0, Appendix A and Math Handout
2	Continuum fluid mechanics and Stress tensor Math review	Appendix A and Math Handout
3	Newton's law of viscosity and the combined momentum flux Math review	Chapter 1
4-5	Introduction to shell momentum balance Flow of a falling film	Chapter 2
6	Flow through a circular tube and modified pressure Creeping flow around a sphere	Chapter 2
7	The equation of continuity and the equations of motion Drifting time derivative and Reynolds transport theorem	Chapter 3
7-8	Solving flow problems using the equations of change	Chapter 3
9-10	Unsteady flow and the method of combination of variables Unsteady flow and method of separation of variables	Chapter 4
11	(March 23-27) Spring break. No classes.	
12	Introduction to turbulent flow and time-smoothed equations of change	Chapter 5
13-14	Friction factors for flow in tubes	Chapter 6
14	Friction factors for flow around submerged objects Pressure drop in packed beds	Chapter 6
15	Macroscopic momentum and mechanical energy balance	Chapter 7
16	Review and problem-solving	